Module 08 Lab Exercise - Contribution Journal

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I completed and ran my own copy of the Module 08 Bias-Variance Tradeoff lab. I created a synthetic sine wave dataset with random noise and plotted the data so I could see the underlying pattern.

I trained polynomial regression models with degrees 1, 4, and 15. I plotted each model's predictions and compared the results. The degree 1 model underfit the data, the degree 4 model provided a good fit, and the degree 15 model overfit the data by creating a very uneven line that followed the individual data points too closely.

I also created a learning curve function using scikit-learn's learning_curve method. I used five-fold cross-validation to compare the training and validation scores for the underfitting and overfitting models. I completed the knowledge check by explaining how the different polynomial degrees represented underfitting, overfitting, and a good fit.